

Assignment 1 (50 marks)

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Deadline: 02/12 11:59 pm

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Problem 1: SVM Classification (35 marks)

lab01_dataset_1.xlsx contains the claim history of 27,513 homeowner policies. The following table describes the eleven columns in the dataset.

Name	Description
policy	Policy Identifier
exposure	Duration a Policy exposed in a Year
num_claims	Number of Claims in a Year
amt_claims	Total Claim Amount in a Year
f_primary_age_tier	Age Tier of Primary Insured
f_primary_gender	Gender of Primary Insured
f_marital	Marital Status of Primary Insured
f_residence_location	Location of Residence Property
f_fire_alarm_type	Fire Alarm Type
f_mile_fire_station	Distance to Nearest Fire Station
f_aoi_tier	Amount of Insurance Tier

We want to predict the *Frequency* which is the *number of claims per unit of exposure* using the above features. We first divide the reported number of claims by the exposure. This gives us the *Frequency*. Next, we put the policies into five groups according to their *Frequency* values. We will use this *Group* as our target variable which has five classes.

Group	Values
0	Frequency = 0
1	$0 < \text{Frequency} \leq 1$
2	$1 < \text{Frequency} \leq 2$
3	$2 < \text{Frequency} \leq 3$
4	$3 < \text{Frequency}$

1.a (6 marks)

Create a new column for the dataset which will indicate the *Frequency Group* and output the updated dataset.

1.b (6 marks)

There are seven categorical features in the dataset namely, *f_aoi_tier*, *f_primary_age_tier*, *f_fire_alarm_type*, *f_marital*, *f_mile_fire_station*, *f_primary_gender*, *f_residence_location*. Display all the unique values of these seven categories.

1.c (6 marks)

We will train SVM models using those seven categorical features. However, their values are currently all categorical data, but SVM requires them to be numerical. Perform **one-hot encoding** on these features to obtain an updated dataset which has only numerical values.

1.d (6 marks)

Divide the observations into training and testing partitions. Observations whose *Policy Identifier* starts with the letters A, G, and P will go to the training partition. The remaining observations go to the testing partition. Output the total number of policies present in the Training partition and Testing partition.

1.e (6 marks)

Train an SVM model using [LinearSVC](#). The input features will be the encoded version of the feature set *f_aoi_tier*, *f_primary_age_tier*, *f_fire_alarm_type*, *f_marital*, *f_mile_fire_station*, *f_primary_gender*, *f_residence_location* and the output is the *Frequency Group*. Use `verbose=1` to observe the optimization steps during the training process.

1.f (5 marks)

Compute and output the accuracy score on the testing partition.

Problem 2: Perceptron Learning (15 marks)

The dataset `lab01_dataset_2.csv` has a *3-dimensional input space* and a class label of *Positive* and *Negative*. For this task, you are **not allowed** to use any *functionalities* of the `sklearn` module.

2.a (10 marks)

Write a function `my_perceptron()` which applies the perceptron algorithm (refer to the lecture slide covering linear separators for details of this algorithm) on the dataset to create a linear separator. `my_perceptron()` takes the dataset as its input and returns a ***3-dimensional weight vector*** which can be used to create the linear separator (assume `bias = 0`). Use the *initial weights* `w = [1, 2, 3]`. Use a classification threshold of 99% i.e., `my_perceptron()` will terminate once the misclassification rate is less than 1%.

2.b (5 marks)

Create a *3D plot* which showcases the dataset in a 3D-space along with the *linear separator* you obtained from `my_perceptron()`. Use two different colors to represent the data points belonging in the two classes for ease of viewing.

Assignment 2 (70 marks)

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Deadline: 03/09 11:59 pm

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Problem 1: Classification (35 marks)

`lab02_dataset_1.xlsx` contains 10,302 observations on various vehicles. You will use the observations in this dataset to train models that predict the usage of a vehicle. The input features which will be used for your training are *AGE*, *TRAVTIME*, *CAR_TYPE*, *OCCUPATION*, *EDUCATION* and your output label is the binary class *CAR_USE*, whose values are *Private* and *Commercial*.

1.a (2 marks)

For the 5 input features, drop any rows with missing values. Output the new length of the training dataset after you drop the missing values among the input features.

1.b (6 marks)

We want to encode categorical features as an integer array. Look up which sklearn function will allow you to do that and apply that on the input features. Also, remember to standardize your encoded features using `fit_transform` to create your final dataset. Finally, perform a 80-20 split on your dataset (80% for training, 20% for testing).

1.c (6 marks)

- Train a logistic regression model using sklearn's `LogisticRegression`.
- Train a naive bayes model using sklearn's `CategoricalNB` with a laplace smoothing of 0.5.

1.d (6 marks)

Compute and output the `accuracy_score` for both the classification models you learned on the testing partition.

1.e (6 marks)

Let us study a few fictitious persons (test cases). Use the `CategoricalNB` model for this task. Person 1, whose age is 45, has a travel time of 46, works in a Manager occupation, has an education level of Masters, and owns a Minivan. Person 2, whose age is 51, has a travel time of 32, works in a Clerical occupation, has an education level of Below High Sc, and owns a Pickup. Person 3, whose age is 25, has a travel time of 14, works in a Blue Collar occupation, has an education level of High School, and owns a SUV.

Display a dataframe that captures all the above information, alongwith two additional columns which will display the Car Use probabilities of these three people.

1.f (5 marks)

Generate a histogram of the predicted probabilities of `CAR_USE = Private` using the `CategoricalNB` model. The bin width is 0.05. The vertical axis is the proportion of **all the observations** (all observations imply both the training + testing dataset as it was in 1.b).

1.g (4 marks)

Using a classification threshold of 0.6 i.e., the class with a probability ≥ 0.6 is the correct output, what is the misclassification rate (computed on **all the observations**) of the `CategoricalNB` model?

Problem 2: Linear Regression (20 marks)

`lab02_dataset_2.csv` contains 238 observations on customer purchase history. You will use the observations in this dataset to train models that predict the loyalty score of a customer. The input features which will be used for your training are *age*, *annual_income*, *purchase_amount*, *purchase_frequency* and your output label is *loyalty_score*.

2.a (2 marks)

Use sklearn's `MinMaxScaler` and `fit_transform` to transform the input features. Then apply a train-test split of 80-20 to generate the training and testing sets.

2.b (10 marks)

Complete the empty functions inside the `myLinearRegression` class so that you can be perform linear regression **without using sklearn**. You are free to add additional helper functions if you need them. We will use a learning rate of 0.01 and 1000 training iterations for the learning task.

```
class myLinearRegression():
    def __init__(self, learning_rate, iterations):
        self.learning_rate = learning_rate
        self.iterations = iterations

        # Model fitting. Initialize all the weights and bias with 0.0.
        # Update them using gradient descent.
        # Use the linear regression formulation as shown in the slides:  $W = W - (\text{learning\_rate}/n) * \text{sum}[X(W.X - Y)]$ 
    def fit(self, X, Y):
        pass

    def predict(self, X):
        pass

model = myLinearRegression(learning_rate = 0.01, iterations = 1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

2.c (2 marks)

Compute and output the `mean_squared_error` between the prediction and the true test labels.

2.d (6 marks)

Now using sklearn's `SGDRegressor` and `LinearRegression` learn two corresponding models using the same training set and test it on the test set. Output the `mean_squared_error` of both the models.

Problem 3: Regularization (15 marks)

`lab02_dataset_3.csv` has 6,435 observations pertaining to Walmart sales and employment. The input features for your training task are *Weekly_Sales*, *Holiday_Flag*, *Temperature*, *Fuel_Price*, *CPI* and the output label is *Unemployment*. For this problem, you will be using regression with regularization.

3.a (3 marks)

Use sklearn's `StandardScaler` and `fit_transform` to transform the input features.

3.b (4 marks)

Use sklearn's `ridge_regression` with `alpha=1.5` to compute the coefficients of the linear regression model. What are the two most important features (feature with the largest positive/negative weights are the most relevant)?

Two most important features:

3.c (4 marks)

Compute the Pearson correlation coefficient between the input features and the output.

The correlation coefficient aligns well with the feature importance as seen in 3.b. Yes or No?

3.d (4 marks)

Create a correlation heatmap using the input features and the output.

Assignment 3 (100 marks)

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Deadline: 04/13 11:59 pm

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Problem 1: PCA (20 marks)

lab03_dataset_1.csv contains 205 observations on various vehicles. This is an unsupervised training data. You will use the entire dataset for PCA.

1.a (2 marks)

For the 14 input features, drop any rows with missing numerical values and output the new length of the training dataset.

1.b (6 marks)

Using the sklearn's PCA method, compute all the principal components (PCs) of the normalized dataset. All the PCs capture a fraction of the total variance, output all the variances captured by all the PCs. Write a code snippet that checks all the PCs and selects the top k PCs whose total variance captured is atleast 90%. What did k come out to be?

1.c (3 marks)

Using the top k PCs, apply dimensionality reduction on the normalized dataset to generate and display the transformed dataset which should now have only k columns. Display the output as a DataFrame.

1.d (3 marks)

We learned in class that we can also obtain the PCs using Eigen Decomposition. But before we can do that, we need to generate the covariance matrix of the normalized dataset. Remember, the covariance matrix can be computed by simply performing the dot product of the normalized matrix with itself, divided by its size.

1.e (3 marks)

Use `Eigen Decomposition` on the covariance matrix to compute and output the eigenvalues and eigenvectors.

1.f (3 marks)

Generate the projections of the normalized dataset using the `first k PCs` (use the `k` obtained in 1.b) and display it as a `DataFrame`.

Problem 2: Clustering (30 marks)

`lab03_dataset_2.csv` contains 239 observations with two input features `x1` and `x2`.

`lab03_dataset_3.csv` contains 1440 observations with two input features `x1` and `x2`.

For this task, you will perform various clustering-related operations using the `sklearn clustering` module.

2.a (6 marks)

Using `lab03_dataset_2.csv`, apply `sklearn's SpectralClustering` algorithm on the two-dimensional data and output the resulting clusters using a scatterplot. You will apply `SpectralClustering` over several clusters ranging from cluster-count `K = 2 to 6`. Make sure for every iteration of different cluster-count, your scatterplot should use `K colors` to clearly distinguish the data points belonging in their respective `K clusters`. Also, compute the `Silhouette score` for each of those `K clusters` and plot that score against `K`. Label the plot axes accordingly.

2.b (6 marks)

Repeat 2.a but instead use `sklearn's AgglomerativeClustering` model for learning.

2.c (6 marks)

Using `lab03_dataset_3.csv`, apply `sklearn's GaussianMixture` on the two-dimensional data and output the resulting clusters using a scatterplot. You will apply `GaussianMixture` over several clusters ranging from cluster-count `K = 2 to 6`. Make sure your scatterplot uses

K colors to clearly distinguish the data points belonging in their respective K clusters. Also, compute the Silhouette score for each of those K clusters and plot the Silhouette score against K clusters. Label the plot axes accordingly.

2.d (6 marks)

Repeat 2.c but instead use sklearn's KMeans model for learning.

2.e (6 marks)

The dataset lab03_dataset_3.csv generates 4 concentric rings, so ideally we would want 4 clusters representing the 4 concentric rings. Did the clustering attempts in 3.c and 3.d lead to 4 concentric ring clusters? Explore some other sklearn clustering algorithms to see which one can actually produce 4 clusters corresponding with the 4 concentric rings and display it.

Problem 3: MLP Classification (25 marks)

Using lab03_dataset_4.csv, train a MLP classifier with sklearn's MLPClassifier. The inputs are alcohol, citric_acid, free_sulfur_dioxide, residual_sugar, sulphates and the output is quality_grp, which has two categories, 0 and 1.

3.a (15 marks)

1. Normalize the input features.
2. Use a train-test split of 80-20.
3. For the learning task, you will train neural network models with different architectures:
Activation function = [logistic, relu, tanh] Hidden layers = [1, 2, 3, 4, 5] Neurons per layer = [2, 4, 6, 8]

So, in the first iteration you will create a learning model using the neural network architecture [logistic, 1, 2], in the second iteration you will use [logistic, 1, 4], all the way to [tanh, 5, 8]. For each of these learned models, compute the misclassification rate on the test set. Once you are done with all the training, you should generate a DataFrame with columns Activation function, Hidden layers, Neurons per layer, Misclassification rate, where each row will correspond with the individual training models. Since the total count of activation functions, hidden layers, neurons are 3, 5, 4 respectively, the number of rows in your output DataFrame should be $3 \times 5 \times 4 = 60$.

3.b (5 marks)

The model with the lowest misclassification rate is the best neural network. Output the model parameters (activation function, hidden layers, neurons) of the best neural network. In the case of ties, choose the network with fewer neurons overall.

3.c (5 marks)

Create a scatterplot of the misclassification rates vs activation functions. The scatterplot should have 3 colors differentiating the misclassification rates associated with the 3 activation functions.

Problem 4: MLP Regression (25 marks)

Using `lab03_dataset_5.csv`, train a MLP classifier with sklearn's `MLPRegressor`. The inputs are `housing_median_age`, `total_rooms`, `households`, `median_income` and the output is `median_house_value`.

4.a (15 marks)

1. Normalize the entire dataset.
2. Use a train-test split of 80-20.
3. For the learning task, you will train neural network models with different architectures: Activation function = [logistic, relu, tanh] Hidden layers = [1, 2, 3, 4, 5] Neurons per layer = [2, 4, 6, 8]

So, in the first iteration you will create a learning model using the neural network architecture [logistic, 1, 2], in the second iteration you will use [logistic, 1, 4], all the way to [tanh, 5, 8]. For each of these learned models, compute the root mean squared errors on the test set. Once you are done with all the training, you should generate a DataFrame with columns Activation function, Hidden layers, Neurons per layer, Root mean squared error, where each row will correspond with the individual training models. Since the total count of activation functions, hidden layers, neurons are 3, 5, 4 respectively, the number of rows in your output DataFrame should be $3 \times 5 \times 4 = 60$.

4.b (5 marks)

The model with the lowest RMSE is the best neural network. Output the model parameters (activation function, hidden layers, neurons) of the best neural network. In the case of ties, choose the network with fewer neurons overall.

4.c (5 marks)

Create a scatterplot of the root mean squared errors vs activation functions. The scatterplot should have 3 colors differentiating the misclassification rates associated with the 3 activation functions.

CS 584 – Machine Learning

Spring 2025

For the course project, your team will develop and deploy a machine learning application with a fully functional frontend and backend. You are free to choose the application area, but you must get it approved before you initiate the project. I will be sharing some projects from past semesters to give you a better understanding of what is expected. All submissions are to be done via Canvas.

- Requires a team of students, ideally 4, but NOT less than 3.
- Each team will maintain a single MS PowerPoint file which will capture the trajectory of their project.
 - Submit a ~5-page project proposal in ppt.
 - Submit a mid-semester update by adding another ~10 pages to the ppt.
 - Submit the final project update by adding another ~10 pages to the ppt.
 - Demo your project to show that it works and submit a final report.
 - Each student in the team will be graded equally unless it is reported that some students did not contribute to the project equally.

Here are the breakdowns pertaining to each of the deliverables discussed above –

- Phase I:
 - State the objective of the project in a clear and comprehensive manner.
 - Briefly discuss background information about the area of your work.
 - Include a rough blueprint/timeline of the project.
 - Have a little discussion about the background research you did so far which can include dataset collections, data preprocessing, etc.
 - Mention the challenges you faced for phase I and how you overcame it.
 - State the next immediate steps you will take for the next phase.
 - Identify some potential challenges that you expect moving forward and potential plans to deal with them.
- Phase II:
 - Add 10 or more slides worth of new materials in the same ppt file which you used for project proposal.
 - Describe the experiments you have done so far with your reasoning to choose specific ML algorithms. Ideally, you will start off with simpler

algorithms to generate a baseline model, before you explore more complex algorithms.

- Show the results of what you have done in phase II, include interesting visualizations such as plots/diagrams/graphs/tables to illustrate your results better.
- Discuss some challenges you faced in this phase and how you had to resolve them.
- Identify some issues that needs to be resolved for the final phase.
- Show a blueprint of your next steps for the final phase.
- Phase III:
 - Add 10 or more slides worth of new materials in the same ppt file which you have been using.
 - Describe any new experiments you have done after the previous phase with your reasoning to choose specific ML algorithms. In this stage, you should explore more complex models to improve the overall performance of your application.
 - Show the results of what you have done in phase III, include interesting visualizations such as plots/diagrams/graphs/tables to illustrate your results better.
 - Illustrate some challenges you faced in this phase and how you had to resolve them.
 - Discuss some potential future possibilities of your project.
 - Discuss some of the potential ethical issues pertaining to the development and use of your application. Explore questions surrounding biases in the model, whether the training data was ethically sourced, who might be impacted from such an application, both positively and negatively etc.
- Demo:
 - Run the model in my presence for the live demo purpose.
 - Your demo should have a functional frontend UI that is easy to operate for a non-technical person.
 - Submit the codebase.
 - Make sure to have proper documentation in the form of a README document that has all the relevant details on how to run the program, required dependencies, and any other information needed to execute the backend code.

- Report:
 - Your final report will be a research-paper style document which will describe everything you did for the project over the semester.
 - Your report should use the double-column format and should have no more than 8-pages (excluding References).
 - You will use the [IEEE conference template](#) for your report. Using LaTeX or LaTeX editor such as [overleaf](#) is highly recommended.
 - The main topics that your report must include are Abstract, Introduction, Related Work, Implementation, Experimental Evaluation, Results and Conclusion. You can have more categories, but these are required.
 - Do not forget to cite all your sources under References, including your dataset sources.
 - Each team will make a single submission on in the form of a pdf file.

Listed below are some resources for obtaining ML datasets:

- [Kaggle Datasets](#)
- [UCI ML Repository](#)
- [Microsoft Datasets](#)
- [Google Datasets](#)
- [US Govt. Data](#)
- [EU Open Data](#)
- [India Govt. Data](#)
- [UK Govt. Data](#)
- [Awesome Public Datasets](#)

CS 584

Spring 2025 Midterm

Name: _____

Id no.: _____

Instructions:

1. This exam is closed book, but handwritten cheat sheet is allowed in the form of a single A4 paper.
2. Computers of any kind are not permitted. Calculators are allowed.
3. Write your answer clearly and legibly in the space provided in the exam booklet.
4. The exam duration is 1 hr. 15 mins.

Final Score (out of 50):

Short-form answers (25 marks)

1. Give an example of non-parametric ML algorithm. _____
2. What algorithm should we use for learning parameters in logistic regression if:
 - a. Objective function is convex. _____
 - b. Objective function is concave. _____
3. Name a global kernel function. _____
4. If events A and B are conditionally independent, given C , what is the probability of B , given both A and C . _____
5. Naïve Bayes assumes conditional independence of all input features. Name the improved Bayesian model that instead assumes the conditional independence over a set of inputs and outputs. _____
6. Name one methodology of reducing overfitting. _____
7. Name the initialization technique that is best suited while dealing with well-separated clusters. _____
8. For binary logistic regression, the functional form of its probability distribution is called a sigmoid. Name the functional form of the probability distribution for multinomial logistic regression. _____
9. Name a loss function that can be optimized for linear regression. _____
10. Name a metric that can be used to measure the cluster quality. _____
11. Given true positives (tp), true negatives (tn), false positives (fp), false negative (fn), recall (R), and precision (P). Write the formula for the following:
 - a. False Positive Rate. _____
 - b. F_2 measure. _____
12. What would you call the phenomenon when learning models suffer from:
 - a. high bias and low variance. _____
 - b. low bias and high variance. _____
13. Answer the following questions with True or False:
 - a. Linear SVM cannot be used for multi-class classification. _____
 - b. For linear SVM, the quadratic optimization problem tries to minimize the distance between 2 hyperplanes. _____
 - c. Complete Linkage uses the smallest distance between all pairs of data points in two clusters. _____
 - d. GMM clustering employs generative learning. _____
 - e. For a dataset with q -dimensional input features and a total of p training instances, the number of parameters we need to learn for linear regression is $(p + 1)$. _____

- f. Smaller learning rate leads to large residual error due to oscillations. _____
- g. Ridge regression uses Euclidean norm. _____
- h. Larger R^2 metric means the model fits the data better. _____
- i. Both AIC and BIC penalize models with large number of parameters relative to the size of the data, but AIC penalizes more severely. _____
- j. Vapnik loss function produces zero loss for small errors. _____

Long-form answers (25 marks)

1. Using perceptron algorithm, derive the linear separator for the given dataset. (5 marks)

Inputs	Output Class
(2, 1)	+
(-2, 3)	+
(-4, 0)	-
(1, -2)	-
(1, 2)	+
(-3, -2)	-

2. Using Naïve Bayes learning on the following dataset, predict the output class (along with its probability) for the test case: <Young, Male, High, Unsafe>. Use Laplace smoothing with $\alpha = 1$ for computing all the likelihoods. (5 marks)

Age	Sex	BP	Cholesterol	Output Class
Young	Female	High	Safe	Drug A
Young	Female	High	Unsafe	Drug A
Middle-age	Female	High	Safe	Drug B
Senior	Female	Normal	Safe	Drug B
Senior	Male	Low	Safe	Drug B
Senior	Male	Low	Unsafe	Drug A
Middle-age	Male	Low	Unsafe	Drug B
Young	Female	Normal	Safe	Drug A
Young	Male	Low	Safe	Drug B
Senior	Male	Normal	Safe	Drug B
Young	Male	Normal	Unsafe	Drug B
Middle-age	Female	Normal	Unsafe	Drug B
Middle-age	Male	High	Safe	Drug B
Senior	Female	Normal	Unsafe	Drug A

3. Apply the Naïve Bayes text classification to the dataset to determine the output class of the document: “winter is not pleasant” from the 3 possible class labels. (5 marks)

Documents	Labels
this place is not ideal	Negative
the weather is hot	Negative
this winter is pleasant	Positive
the ideal place is beach	Positive
sunny weather is ideal for beach	Neutral
winter is coming	Neutral

4. Using linear regression, compute the parameters obtained after one iteration of the gradient descent algorithm, assuming the learning rate is 1.0 and the current parameters are $[w_1, w_2, w_3] = [-0.3, 0.7, 0.5]$. Ignore the bias in this computation, so you only need to update 3 parameters. With the newly obtained weights, compute the RMSE. (10 marks)

x_1	x_2	x_3	y
-1	-3	2	-1.3
0	2	-1	1.7
3	-1	1	2.5

CS 584

Spring 2025 Final

Name: _____

Id no.: _____

Instructions:

1. This exam is closed book, but handwritten cheat sheet is allowed in the form of a single A4 paper.
2. Computers of any kind are not permitted. Calculators are allowed.
3. Write your answer clearly and legibly in the space provided in the exam booklet.
4. The exam duration is 2 hrs.

Final Score (out of 100):

Short-form answers (20×2 = 40 marks)

1. What is the name of the equation that an optimal value function for reinforcement learning algorithm must satisfy? _____
2. What is the name of the loss function which gives zero loss for small errors and absolute loss for larger errors? _____
3. Name an on-policy reinforcement learning algorithm. _____
4. Which LSTM gate decides how much long-term memory passes on? _____
5. Name the technique that uses a deep learning algorithm to label each pixel in an image with a class label. _____
6. What kind of ReLU is used to counter the dying ReLU problem? _____
7. Let B and V be the bias and variance of a learning model. What is the expected loss in terms of B and V? _____
8. For a CNN with an input shape of (128, 128, 3), what are the number of parameters for a convolution layer with 5 filters of shape (10, 10, 3), no padding and stride of 2? _____
9. What will be the shape of the output after the above convolution? _____
10. LSTM cells have 3 gates, how many gates does GRU cells have? _____
11. Name a density-based clustering technique. _____
12. Name a two-stage object detection algorithm. _____
13. What is the sum of each row of an attention matrix in a transformer? _____
14. What is the name of the RL technique which can learn a parameterized policy that can select actions without consulting a value function? _____
15. Name the technique which is used to train an RNN via updating its network weights. _____
16. Name the technique used in object detection to eliminate duplicate overlapping bounding boxes. _____
17. In value learning RL, name the value function being maximized. _____
18. What is the component that is added in SVM to allow some misclassifications thereby leading to soft margin classification? _____
19. In a transformer neural network, if the query, key, and value matrices are each of size $p \times q$, what is the size of the self-attention matrix. _____
20. Name the technique where the weights in an ANN are multiplied by some fraction in (0, 1) after every epoch. _____

True/False answers (10 marks)

1. PCA can be computed via minimizing reconstruction error. _____
2. Kernel PCA is an example of linear PCA. _____

3. Vanilla RNN suffers from the vanishing gradient problem. _____
4. Adaptive learning rates can lead to faster convergence. _____
5. VGGNet uses skip connection in its architecture. _____
6. All principal components are orthogonal to one another. _____
7. Feature maps become denser as we go deeper in CNN. _____
8. Silhouette score is not a good metric to capture the quality of clustering for non-globular clusters. _____
9. Seq2Seq uses the encoder-decoder architecture. _____
10. k-fold cross validation technique is more computationally expensive than LOOCV. _____

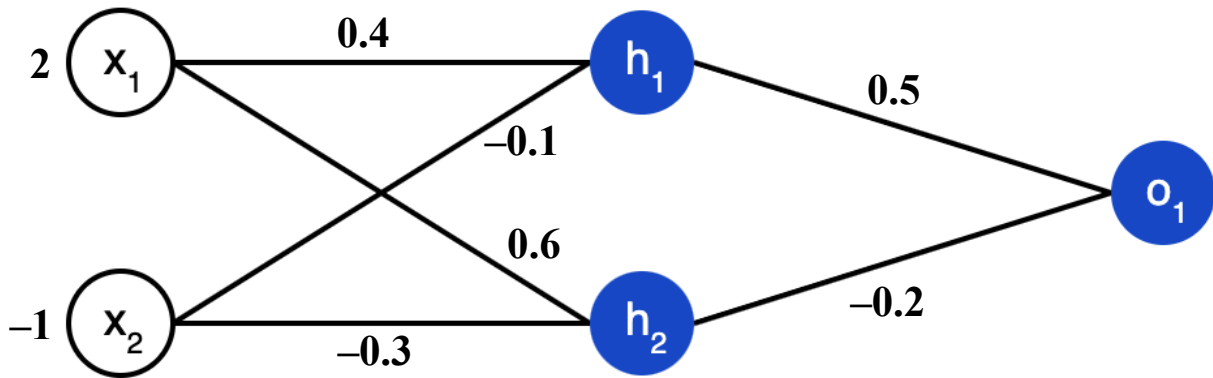
Long-form answers (50 marks)

1. Convolve the 6×4 image array with the 2×2 filter (stride = 2). Then, apply a ReLU activation on the convolved image, followed by average pooling with a 1×2 filter (stride = 1). (10 marks)

20	5	70	200
30	0	0	100
70	60	210	0
170	160	5	1
90	0	10	100
1	20	80	5

-1	3
-2	2

2. For the following MLP with sigmoid activation on its hidden and output layers, given the input values and weights, generate all the network weights learned after one iteration of backpropagation. Assume no bias, learning rate is 1, and the true output is 1. (10 marks)



3. Given an LSTM cell with input $x^t = -2.5$, hidden state $h^{t-1} = -0.8$, and cell state $c^{t-1} = 1.2$, generate the outputs h^t and c^t , if the internal weights and biases are as follows: $w_{xf} = 0.1$, $w_{hf} = 2.3$, $b_f = 2.1$, $w_{xi} = -1.4$, $w_{hi} = -1.1$, $b_i = -1.7$, $w_{xc} = 0.9$, $w_{hc} = -0.5$, $b_c = 1.8$, $w_{xo} = -0.8$, $w_{ho} = 1.6$, $b_o = -2.6$ (10 marks)

4. Using the training dataset below, apply the Naïve Bayes text classification to determine the output class of the document: “the movie theater is costly” from the 3 possible class labels. Use Laplace smoothing with $\alpha = 1$. (10 marks)

Documents	Labels
this movie is amazing	Positive
the movie price is very costly	Negative
this theater is very clean	Positive
the food at this theater is not costly	Neutral
afternoon is ideal for this movie	Neutral
the theater food is not amazing	Negative
clean theater with amazing movie is ideal	Positive
theater is very crowded for this movie	Negative

5. Using Naïve Bayes learning on the following dataset, predict the output class (Can we pet them = Yes/No), along with its probability for the following test cases. Use Laplace smoothing with $\alpha = 1$. (10 marks)
- <Dog, Small, White>
 - <Cow, Medium, Brown>

Animals	Size of Animal	Body Color	Can we Pet them
Dog	Medium	Black	Yes
Dog	Big	White	No
Rat	Small	White	Yes
Cow	Big	White	Yes
Cow	Small	Brown	No
Cow	Big	Black	Yes
Rat	Big	Brown	No
Dog	Small	Brown	Yes
Dog	Medium	Brown	Yes
Cow	Medium	White	No
Dog	Small	Black	Yes
Rat	Medium	Black	No
Rat	Small	Brown	No
Cow	Big	White	Yes